

EDUCATION

- **The Hong Kong University of Science and Technology** Hong Kong SAR
Ph.D. in Computer Science and Engineering Sep. 2019 – July 2024
 - **Advisor:** Prof. Ke Yi
 - **Thesis:** Secure Multi-Party Query Processing
- **Fudan University** Shanghai, China
B.Sc. in Computer Science and Technology (Honor Class) Sep. 2015 – June 2019

WORK EXPERIENCE

- **Ant Research, Ant Group** Shanghai, China
Associate Researcher Aug. 2024 – Present
 - **Research Focus:** Data security and privacy, database theory and algorithms
- **LeetCode** Shanghai, China
Technical Solution Author and Problem Adder Apr. 2019 – Sep. 2020
- **OceanBase, Ant Group** Hangzhou, China
Software Engineer Intern Apr. 2019 – July 2019

PUBLICATIONS

* indicates equal contribution.

1. Wei Dong, **Qiyao Luo**, Giulia Fanti, Elaine Shi, and Ke Yi. “Instance-optimal Clipping for Summation Problems in the Shuffle Model of Differential Privacy.” *ACM SIGSAG Conference on Computer and Communications Security (CCS)*, Oct. 2024.
2. **Qiyao Luo**, Yilei Wang, Ke Yi, Sheng Wang, and Feifei Li. “Secure Sampling for Approximate Multi-party Query Processing.” *Proceedings of the ACM on Management of Data (SIGMOD)*, June 2024.
3. Qichen Wang*, **Qiyao Luo***, and Yilei Wang*. “Relational Algorithms for Top-k Query Evaluation.” *Proceedings of the ACM on Management of Data (SIGMOD)*, June 2024.
4. Wei Dong, Zijun Chen, **Qiyao Luo**, Elaine Shi, and Ke Yi. “Continual Observation of Joins under Differential Privacy.” *Proceedings of the ACM on Management of Data (SIGMOD)*, June 2024.
5. Wei Dong, **Qiyao Luo**, and Ke Yi. “Continual Observation under User-level Differential Privacy.” *IEEE Symposium on Security and Privacy (S&P)*, May 2023.
6. **Qiyao Luo**, Yilei Wang, and Ke Yi. “Frequency Estimation in the Shuffle Model with Almost a Single Message.” *ACM SIGSAG Conference on Computer and Communications Security (CCS)*, Nov. 2022.

PREPRINTS

1. **Qiyao Luo**, Yilei Wang, Wei Dong, and Ke Yi. “LINQ: Secure Query Processing with Linear Complexity.” 2024.
2. **Qiyao Luo**, Yilei Wang, Zhenghang Ren, Ke Yi, Kai Chen, and Xiao Wang. “Secure Machine Learning over Relational Data.” 2021.

SERVICES

- External Reviewer: Eurocrypt (2025), SIGMOD (2020-2024), PODS (2019, 2024)

COMPETITIONS

- Champion, Gold Medal, ACM-ICPC Asia Hong Kong Regional Contest Nov. 2017
- Gold Medal, ACM-ICPC Asia CHINA-Final Contest Nov. 2016
- Gold Medal, China Collegiate Programming Contest Final Nov. 2016
- First Prize, China Undergraduate Mathematical Contest in Modeling Sep. 2016
- Silver Medal, Chinese Mathematical Olympiad (**CMO**) Jan. 2015
- Copper Medal, National Olympiad in Informatics (**NOI**) Aug. 2014

AWARDS AND SCHOLARSHIPS

- RedBird Academic Excellence Award, HKUST 2023, 2024
- Postgraduate Scholarship, HKUST 2019 – 2023
- Outstanding Honor Class Students, Fudan University 2017, 2018
- First Class Scholarship, Fudan University 2016
- Freshman Scholarship, Fudan University 2015

SKILLS

- **Programming:** C/C++, Python, Spark, SQL, and LaTeX.
- **Languages:** Mandarin, English, and Cantonese.